# Smart Warehouse Management System

## Use Case Descriptions

Phase 1 — Software Requirements Specification

CS251 Software Engineering 1 | Spring 2025-2026

## Use Case Descriptions — All 20 Use Cases

.

### Warehouse Manager Use Cases

#### UC-01 — View Inventory Health Dashboard

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| **Use Case ID** | UC-01 |
| **Use Case Name** | View Inventory Health Dashboard |
| **Actor(s)** | Warehouse Manager |
| **Description** | The manager views a real-time dashboard showing inventory status, IoT sensor readings, stock discrepancies, and environmental alerts. |
| **Preconditions** | 1. Manager is logged in.<br>2. Inventory data is available in the system. |
| **Main Flow** | 1. Manager opens the Inventory Dashboard.<br>2. System retrieves current stock levels for all zones.<br>3. System displays IoT weight sensor readings (Function 3).<br>4. System highlights any stock discrepancies.<br>5. System shows environmental alerts for sensitive items (Function 5).<br>6. Manager reviews the dashboard. |
| **Extension Points** | capacity_alert: triggered if warehouse occupancy > 90%.<br>expiry_check: triggered if any item expiry date is within 7 days. |
| **Postconditions** | 1. Dashboard is displayed with up-to-date inventory data.<br>2. Any alerts are visible to the manager. |

#### UC-02 — Manage Zonal Storage

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| **Use Case ID** | UC-02 |
| **Use Case Name** | Manage Zonal Storage |
| **Actor(s)** | Warehouse Manager |
| **Description** | The manager uses the zonal storage optimizer to assign items to appropriate zones based on velocity and dimensions, and monitors zone capacity. |
| **Preconditions** | 1. Manager is logged in.<br>2. Item catalog and zone data are available. |
| **Main Flow** | 1. Manager opens the Zonal Storage Management screen.<br>2. System runs the Smart Zonal Storage Optimizer (Function 1).<br>3. System suggests optimal storage locations based on item velocity.<br>4. System calculates 3D volumetric capacity per zone (Function 2).<br>5. Manager approves or overrides suggested locations.<br>6. System updates zone assignments. |
| **Extension Points** | BackOrder: triggered if an incoming item has an existing backorder (routes to Cross-Docking). |
| **Postconditions** | 1. Items are assigned to optimal zones.<br>2. Zone capacity is updated in the system. |

## UC-03 — Monitor Perishable & Expiry Items

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| **Use Case ID** | UC-03 |
| **Use Case Name** | Monitor Perishable & Expiry Items |
| **Actor(s)** | Warehouse Manager |
| **Description** | The system flags items nearing their Best Before date and suggests FEFO picking order. |
| **Preconditions** | 1. Manager is logged in.<br>2. Items with expiry dates exist in the system. |
| **Main Flow** | 1. System scans all items with expiry dates (Function 4).<br>2. System identifies items expiring within 7 days.<br>3. System flags these items on the dashboard.<br>4. System suggests First-Expiry, First-Out (FEFO) picking order.<br>5. Manager reviews flagged items and confirms FEFO instructions. |
| **Extends** | UC-01 (at extension point: expiry_check) |
| **Condition** | {if item expiry date <= 7 days from today} |
| **Postconditions** | 1. Expiring items are flagged.<br>2. FEFO picking instructions are sent to Floor Staff. |

## UC-04 — View Supplier Performance & Analytics

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|---|---|
| **Use Case ID** | UC-04 |
| **Use Case Name** | View Supplier Performance & Analytics |
| **Actor(s)** | Warehouse Manager |
| **Description** | The manager reviews supplier accuracy scores, delivery performance, and contract terms before making procurement decisions. |
| **Preconditions** | 1. Manager is logged in.<br>2. Historical supplier data exists in the system. |
| **Main Flow** | 1. Manager selects 'Evaluate before order' option.<br>2. System calculates Supplier Accuracy Score (Function 15).<br>3. System displays ratio of items ordered vs. correctly received.<br>4. System shows supplier contract versions and pricing tiers (Function 22).<br>5. Manager reviews and selects preferred supplier. |
| **Extends** | UC-05 (at extension point: Evaluation) |
| **Condition** | {if manager selects 'evaluate before order'} |
| **Postconditions** | 1. Supplier performance report is displayed.<br>2. Manager decision is recorded. |

### UC-05 — Manage Procurement & Purchase Orders

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|---|---|
| **Use Case ID** | UC-05 |
| **Use Case Name** | Manage Procurement & Purchase Orders |
| **Actor(s)** | Warehouse Manager |
| **Description** | The manager oversees the procurement process including stock reorder triggers, supplier selection, and purchase order generation. |
| **Preconditions** | 1. Manager is logged in.<br>2. Stock levels are available.<br>3. Supplier data is configured. |
| **Main Flow** | 1. System detects stock below Safety Stock threshold (Function 12).<br>2. System sends automated notification to the Manager.<br>3. Manager reviews reorder recommendation.<br>4. System applies Tiered Supplier Selection Logic (Function 16).<br>5. System generates B2B Purchase Order with itemized SKUs (Function 14).<br>6. Manager approves and sends PO to Supplier. |
| **Extension Points** | Evaluation: triggered if manager selects 'evaluate before order'. |
| **Includes** | UC-16 (Receive & Confirm Purchase Order) |
| **Postconditions** | 1. Purchase Order is generated and sent to Supplier.<br>2. Stock reorder is logged. |

### UC-06 — View Warehouse Capacity Alerts

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|---|---|
| **Use Case ID** | UC-06 |
| **Use Case Name** | View Warehouse Capacity Alerts |
| **Actor(s)** | Warehouse Manager |
| **Description** | The system automatically notifies the manager when total warehouse occupancy exceeds 90% using the Observer Pattern. |
| **Preconditions** | 1. Manager is logged in.<br>2. Occupancy monitoring is active. |
| **Main Flow** | 1. System continuously monitors total warehouse occupancy (Function 39).<br>2. Occupancy exceeds 90%.<br>3. Observer Pattern triggers an automatic alert.<br>4. System displays alert on the Manager's dashboard.<br>5. Manager reviews occupancy report and takes action. |
| **Extends** | UC-01 (at extension point: capacity_alert) |
| **Condition** | {if warehouse occupancy > 90%} |
| **Postconditions** | 1. Capacity alert is displayed.<br>2. Alert is logged in the system. |

## UC-07 — Activate Emergency Mode

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|---|---|
| **Use Case ID** | UC-07 |
| **Use Case Name** | Activate Emergency Mode |
| **Actor(s)** | Warehouse Manager |
| **Description** | The manager activates Emergency Mode to immediately pause all picking tasks and lock dock doors during a security breach. |
| **Preconditions** | 1. Manager is logged in.<br>2. A security breach or emergency is detected. |
| **Main Flow** | 1. Manager clicks the 'Activate Emergency Mode' button (Function 35).<br>2. System pauses all active picking tasks for Floor Staff.<br>3. System locks all dock doors.<br>4. System sends emergency notification to all active users.<br>5. Manager monitors the situation from the dashboard. |
| **Postconditions** | 1. All picking tasks are paused.<br>2. Dock doors are locked.<br>3. Emergency state is logged. |

## UC-08 — Manage User Roles & Access

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|---|---|
| **Use Case ID** | UC-08 |
| **Use Case Name** | Manage User Roles & Access |
| **Actor(s)** | Warehouse Manager |
| **Description** | The manager assigns and manages role-based access control, ensuring Floor Staff can only access picking/packing screens. |
| **Preconditions** | 1. Manager is logged in.<br>2. User accounts exist in the system. |
| **Main Flow** | 1. Manager opens the User Management screen.<br>2. Manager selects a user account.<br>3. System displays RBAC options (Function 33).<br>4. Manager assigns role: Manager, Picker, Packer, or Supplier.<br>5. System restricts system access based on assigned role.<br>6. Manager saves changes. |
| **Postconditions** | 1. User role is updated.<br>2. Access permissions are applied immediately. |

### UC-09 — Archive & Retain Data

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|---|---|
| **Use Case ID** | UC-09 |
| **Use Case Name** | Archive & Retain Data |
| **Actor(s)** | Warehouse Manager |
| **Description** | The system automatically moves completed order data to long-term storage after 12 months to maintain database performance. |
| **Preconditions** | 1. Manager is logged in.<br>2. Completed orders older than 12 months exist. |
| **Main Flow** | 1. System runs automatic archive check (Function 37).<br>2. System identifies completed orders older than 12 months.<br>3. System moves data to long-term archive storage.<br>4. System notifies Manager that archiving is complete.<br>5. Manager can retrieve archived data if needed. |
| **Postconditions** | 1. Old data is archived.<br>2. Active database performance is maintained. |

## Picker Use Cases

### UC-10 — Perform Batch Picking

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|---|---|
| **Use Case ID** | UC-10 |
| **Use Case Name** | Perform Batch Picking |
| **Actor(s)** | Picker |
| **Description** | The Picker receives an optimized pick list combining multiple orders to minimize walking distance inside the warehouse. |
| **Preconditions** | 1. Picker is logged in.<br>2. Active orders are available for picking.<br>3. No Emergency Mode is active. |
| **Main Flow** | 1. System runs Batch Picking Route Logic (Function 23).<br>2. System combines multiple orders into a single Pick List.<br>3. System calculates optimal walking route to minimize distance.<br>4. Picker receives the Pick List on their device.<br>5. Picker walks the route and scans items.<br>6. System updates Order Fulfillment Status (UC-14). |
| **Includes** | UC-14 (Track Order Fulfillment Status) |
| **Postconditions** | 1. All items in the Pick List are collected.<br>2. Order status is updated to 'Picking Complete'. |

## Packer Use Cases

### UC-11 — Pack & Sort Orders

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|---|---|
| **Use Case ID** | UC-11 |
| **Use Case Name** | Pack & Sort Orders |
| **Actor(s)** | Packer |
| **Description** | The Packer sorts picked items into customer-specific bins, selects optimal box size, validates weight, and generates a shipping label. |
| **Preconditions** | 1. Packer is logged in.<br>2. Picked items have arrived at the packing station. |
| **Main Flow** | 1. System guides Packer via Sort-to-Light simulation (Function 25).<br>2. Packer places items into correct customer bins.<br>3. System suggests smallest box size (Function 30).<br>4. Packer packs the items.<br>5. System validates parcel weight and dimensions (UC-12).<br>6. System generates shipping label (UC-13). |
| **Includes** | UC-12 (Validate Parcel Weight & Dimensions), UC-13 (Generate & Scan Shipping Label) |
| **Postconditions** | 1. Order is packed and labeled.<br>2. Order status updated to 'Packed'. |

### UC-12 — Validate Parcel Weight & Dimensions

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|---|---|
| **Use Case ID** | UC-12 |
| **Use Case Name** | Validate Parcel Weight & Dimensions |
| **Actor(s)** | Packer |
| **Description** | The system compares the final packed box weight against the expected sum of item weights to detect packing errors. |
| **Preconditions** | 1. Packer has finished packing an order.<br>2. Item weight data is available in the system. |
| **Main Flow** | 1. Packer places packed box on the scale.<br>2. System reads the actual box weight (Function 26).<br>3. System calculates the expected weight from item records.<br>4. System compares actual vs. expected weight.<br>5a. If match: System approves the parcel.<br>5b. If mismatch: System alerts the Packer to recheck contents. |
| **Includes** | Part of UC-11 (Pack & Sort Orders) |
| **Postconditions** | 1. Parcel weight is validated.<br>2. Packing error (if any) is flagged. |

### UC-13 — Generate & Scan Shipping Label

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|---|---|
| **Use Case ID** | UC-13 |
| **Use Case Name** | Generate & Scan Shipping Label |
| **Actor(s)** | Packer |
| **Description** | The system creates a unique scannable shipping label linking the physical box to the digital order record. |
| **Preconditions** | 1. Order has been packed and validated.<br>2. Shipping carrier has been selected. |
| **Main Flow** | 1. System generates a unique shipping label with QR code (Function 31).<br>2. System links the label to the digital order record.<br>3. Packer prints the label.<br>4. Packer scans the label to confirm attachment.<br>5. System marks order as 'Label Generated'. |
| **Includes** | Part of UC-11 (Pack & Sort Orders) |
| **Postconditions** | 1. Shipping label is generated and scanned.<br>2. Order is linked to the physical box. |

## Floor Staff (Shared) Use Cases

### UC-14 — Track Order Fulfillment Status

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|---|---|
| **Use Case ID** | UC-14 |
| **Use Case Name** | Track Order Fulfillment Status |
| **Actor(s)** | Floor Staff (Picker & Packer) |
| **Description** | Floor Staff tracks the state of an order through the fulfillment state machine: Processing → Picking → Packing → Shipped → Delivered. |
| **Preconditions** | 1. Floor Staff is logged in.<br>2. An active order exists in the system. |
| **Main Flow** | 1. System displays current order status (Function 28).<br>2. Staff selects an order to view.<br>3. System shows the state machine: Processing → Picking → Packing → Shipped → Delivered.<br>4. Staff confirms transition to next state.<br>5. System updates the order status automatically. |
| **Postconditions** | 1. Order status is updated.<br>2. Next actor in the workflow is notified. |

### UC-15 — Handle Cross-Docking

| | |
|---|---|
| **Use Case ID** | UC-15 |
| **Use Case Name** | Handle Cross-Docking |
| **Actor(s)** | Picker |
| **Description** | The system identifies incoming items that are already backordered and routes them directly to the Packing station instead of storage. |
| **Preconditions** | 1. An incoming shipment has arrived.<br>2. A backorder exists for items in the shipment. |
| **Main Flow** | 1. System checks incoming items against backorder list (Function 8).<br>2. System identifies backordered items.<br>3. System routes backordered items directly to Packing station.<br>4. Picker is notified to deliver items to Packing, not Storage.<br>5. System updates inventory to reflect cross-docking. |
| **Extends** | UC-02 (at extension point: BackOrder) |
| **Condition** | {if incoming item has existing backorder} |
| **Postconditions** | 1. Backordered items are routed to Packing.<br>2. Backorder record is updated. |

## Supplier Use Cases

### UC-16 — Receive & Confirm Purchase Order

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|---|---|
| **Use Case ID** | UC-16 |
| **Use Case Name** | Receive & Confirm Purchase Order |
| **Actor(s)** | Supplier |
| **Description** | The Supplier receives an automated Purchase Order from the system and confirms receipt. |
| **Preconditions** | 1. Supplier is logged in.<br>2. A Purchase Order has been generated by the Manager. |
| **Main Flow** | 1. System sends PO to Supplier portal (Function 14).<br>2. Supplier reviews PO details: SKUs, quantities, prices.<br>3. Supplier confirms or requests modification.<br>4. System records Supplier confirmation.<br>5. System updates PO status to 'Confirmed'. |
| **Extends** | Part of UC-05 (Manage Procurement) |
| **Condition** | Always triggered when a PO is generated. |
| **Postconditions** | 1. PO is confirmed by Supplier.<br>2. Procurement workflow continues. |

### UC-17 — Update Shipment & Lead Time

| | |
|---|---|
| **Use Case ID** | UC-17 |
| **Use Case Name** | Update Shipment & Lead Time |
| **Actor(s)** | Supplier |
| **Description** | The Supplier updates shipment details and the system estimates arrival based on historical delivery performance. |
| **Preconditions** | 1. Supplier is logged in.<br>2. A confirmed PO exists. |
| **Main Flow** | 1. Supplier updates shipment dispatch details.<br>2. System calculates estimated arrival using Lead-Time Estimator (Function 13).<br>3. System records shipment status: Expected → At Dock → Being Inspected → Stored (Function 18).<br>4. System notifies Manager of estimated arrival.<br>5. System triggers Backorder Management if needed (UC-18). |
| **Extension Points** | Carrier: triggered if no carrier has been assigned to the shipment. |
| **Includes** | UC-18 (Manage Backorders) |
| **Postconditions** | 1. Shipment details updated.<br>2. Estimated arrival communicated to Manager. |

### UC-18 — Manage Backorders

| | |
|---|---|
| **Use Case ID** | UC-18 |
| **Use Case Name** | Manage Backorders |
| **Actor(s)** | Supplier |
| **Description** | The system prioritizes incoming stock for customers who have been waiting the longest when a backordered shipment arrives. |
| **Preconditions** | 1. A shipment containing backordered items has arrived.<br>2. Backorder records exist in the system. |
| **Main Flow** | 1. System detects arrival of backordered items (Function 21).<br>2. System sorts customers by waiting time (longest first).<br>3. System allocates incoming stock to waiting customers.<br>4. System notifies affected customers and Floor Staff.<br>5. System updates backorder records. |
| **Extends** | Part of UC-17 (Update Shipment & Lead Time) |
| **Condition** | Always triggered when backordered items arrive. |
| **Postconditions** | 1. Backorders are fulfilled in priority order.<br>2. Backorder records are updated. |

### UC-19 — Select Shipping Carrier

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|---|---|
| **Use Case ID** | UC-19 |
| **Use Case Name** | Select Shipping Carrier |
| **Actor(s)** | Supplier |
| **Description** | The system selects the best shipping carrier based on the customer's delivery address, balancing speed and cost. |
| **Preconditions** | 1. An order is ready for shipment.<br>2. No carrier has been assigned yet. |
| **Main Flow** | 1. System evaluates shipping options (Function 27).<br>2. System applies carrier selection logic: Fast vs. Cost-effective.<br>3. System considers customer delivery address and urgency.<br>4. System selects optimal carrier.<br>5. Carrier assignment is recorded and label generation begins. |
| **Extends** | UC-17 (at extension point: Carrier) |
| **Condition** | {if no carrier assigned to shipment} |
| **Postconditions** | 1. Shipping carrier is selected.<br>2. Carrier details linked to the shipment. |

### UC-20 — Undergo Performance Audit

| | |
|---|---|
| **Use Case ID** | UC-20 |
| **Use Case Name** | Undergo Performance Audit |
| **Actor(s)** | Supplier |
| **Description** | The system evaluates the Supplier's performance by calculating an accuracy score based on order fulfillment history. |
| **Preconditions** | 1. Supplier is logged in.<br>2. Historical order data exists for the Supplier. |
| **Main Flow** | 1. System retrieves Supplier's delivery history (Function 15).<br>2. System calculates Accuracy Score: items correctly received / items ordered.<br>3. System displays performance report to Supplier.<br>4. System updates Supplier ranking in Tiered Selection Logic (Function 16).<br>5. Supplier reviews performance report. |
| **Postconditions** | 1. Supplier Accuracy Score is updated.<br>2. Supplier tier ranking is adjusted. |